



Society of Petroleum Engineers

SPE-228894-MS

Nanotechnology Sealant Reduces Filtrate Invasion, Stabilizes Shale and Coal Formations in East Kalimantan Onshore

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Copyright 2025, Society of Petroleum Engineers DOI [10.2118/228894-MS](https://doi.org/10.2118/228894-MS)

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

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Abstract

In mature oil and gas fields, accessing hydrocarbons efficiently can be challenging, especially in depleted reservoirs where minor permeability disruptions can cause significant formation damage, impacting economic viability. A major operator in East Kalimantan Indonesia aimed to replace non-aqueous drilling fluids with cost-effective, environmentally friendly water-based alternatives. This system sought to stabilize shale and coal formations while minimizing reservoir damage through reduced fluid invasion, enhancing operational success and economic value.

The reservoir in this area consists of sandstone, shale, and coal layers, each requiring specific treatment without compromising the others. Laboratory tests and software simulations were conducted to develop an optimal drilling fluid system. Shale studies, including X-ray diffraction, determined the right fluid system, while proprietary software simulated a bridging package to match the reservoir pore throats. Formation response tests using core plugs confirmed reservoir compatibility. The Nano high-performance water-based fluid (NHPWBF) was then selected, meeting criteria for shale stabilization with zero-spurt loss and low total filtrate, ensuring minimal formation damage and improved wellbore stability.

The NHPWBF system, incorporating an advanced nano sealant, outperformed other systems in laboratory testing. Based on these results, the NHPWBF system was implemented in the "M" field, with regular High-Performance Water-Based Fluid (HPWBF) used in the overburden section and NHPWBF in the reservoir. Drilling proceeded without shale or clay issues, firm cuttings indicated good shale inhibition, coal layers were drilled without any problems, and drilling was completed without abnormal torque or drag. Drilling fluid cost was better optimized, thus contributing to reducing the overall cost compared to planned authorization for expenditure (AFE). Wireline logging achieved targets efficiently, with sampling times below the average time from previous wells. Pressure testing and fluid sampling were successful, with no filtrate detected, confirming the NHPWBF's sealing capability and creation of a non-permeable filter cake. Cost savings greater than 30% were achieved when compared to similar wells drilled with non-aqueous fluids, and well testing demonstrated natural flow, a rarity in the area. Two wells have been successfully

completed using this fluid system, showing consistent performance and results, and creating an opportunity to expand the NHPWBFs use across the field for improved efficiency and cost-effectiveness.

The addition of nano sealant technology in HPWBF enhanced filter cake quality, creating a relatively impermeable layer that stabilized the wellbore through reduced pore pressure transmission and improved shale/clay stability. It also stabilized coal formations and minimized filtrate invasion, reducing reservoir damage. NHPWBF improved drilling performance, improved economics, and set a new fluid system standard in the area.

Introduction

As one of the main sources of energy, oil and gas are vital products that need to be produced continuously to ensure a steady supply to the world. Oil and gas industry is a complex operation that involves multiple disciplines in sciences and technology that stretch from upstream to downstream. The drilling operation involves activities that are both labor- and capital-intensive but very necessary to ensure the availability of oil and gas.

The "M" field, in East Kalimantan is a mature field that has been under development for over 50 years. Drilling operations have been well-established and standardized in this region. A typical well design here involves two sections: the top-hole section drilled with a 12.25-inch bit using HPWBF, and the reservoir section drilled with an 8.5-inch bit using Synthetic Oil Based Mud (SOBM).

Between 2020 and 2021, the Operator replaced SOBM with HPWBF for drilling from the top-hole through to the reservoir section, achieving favorable drilling performance. However, subsequent evaluation by the subsurface team revealed that filtrate from the HPWBF had penetrated too deeply into the reservoir formation. This excessive fluid invasion is suspected to be a contributing factor to the lower-than-anticipated production rates on several wells. As a result of these findings, the Operator decided to resume the use of SOBM in the reservoir section.

Drilling operations in this region are frequently challenged by depleted reservoir pressures, sloughing shales, fragile formations such as coal layers, and a high risk of formation damage. A key contributor to these issues is excessive fluid invasion. In depleted zones, substantial filtrate—or even whole drilling fluid—can infiltrate deeply into the reservoir, potentially altering its natural characteristics and causing formation damage. In shale formations, excessive fluid invasion can alter the pore pressure and weaken the shale's strength, leading to wellbore instability¹. Unlike SOBM, where filtrate invasion is typically minimal due to higher capillary entry pressures, water-based fluid (WBF) filtrates tend to penetrate more extensively, exacerbating instability². The filtrate that invades the shale matrix, elevating the near-wellbore pore pressure over time and reducing the effective stress, ultimately results in shear failure². A similar mechanism contributes to coal layer instability, where fluid invasion into the coal cleats system causes pressure buildup and diminishes mud weight support, leading to structural failure³.

With all these problems, SOBM seems the best solution to effectively provide wellbore stability, good shale inhibition and is expected to minimize formation damage with its capability to provide a tight filter cake. However, the use of SOBM in this area introduces high operational costs and environmental concerns. Considering these SOBM limitations, the operator again sought to adopt HPWBF alternatives that could meet both the technical and economic challenges while reducing environmental impacts.

The key challenge was to design a fluid system that could effectively address the diverse geological formations encountered in the region, particularly interbedded sandstone, shale, and coal layers while minimizing formation damage. Each lithology has unique characteristics and reacts differently to fluid systems. The goal was to create a water-based drilling fluid that could provide wellbore stability, limit filtrate invasion, and preserve reservoir productivity.

Geological Background and Well Profile

The "M" Field located in Onshore East Kalimantan, Indonesia comprises a complex lithology of alternating sandstone, shale, and coal layers along the reservoir. The sandstone serves as the primary hydrocarbon-bearing formation, while the shale and coal present stability and permeability management challenges. Drilling through these formations using a single fluid system requires a delicate balance to prevent borehole instability and minimize formation damage.

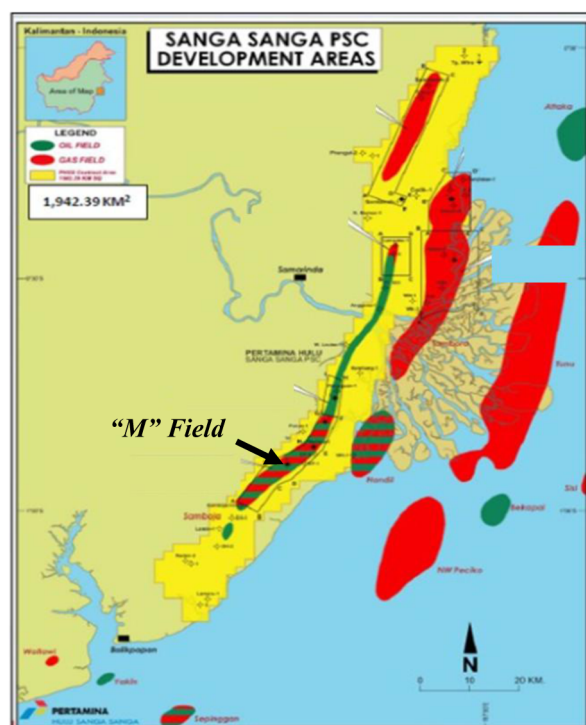


Figure 1—"M" Field location

Shale formations are particularly sensitive to water-based fluids due to their tendency to swell, disperse, or collapse when hydrated. Coal formations, while generally brittle, are prone to instability and fluid loss if not adequately sealed. Therefore, fluid design had to account for inhibitive, sealing, and lubricating properties to handle geological diversity.

Below are the graphs of the pressure/stress profile of the well to be drilled along with the lithology and casing design. Reservoir section in this well consists of Shale, Coal and unconsolidated sandstone formation that is already depleted due to massive production from this reservoir.

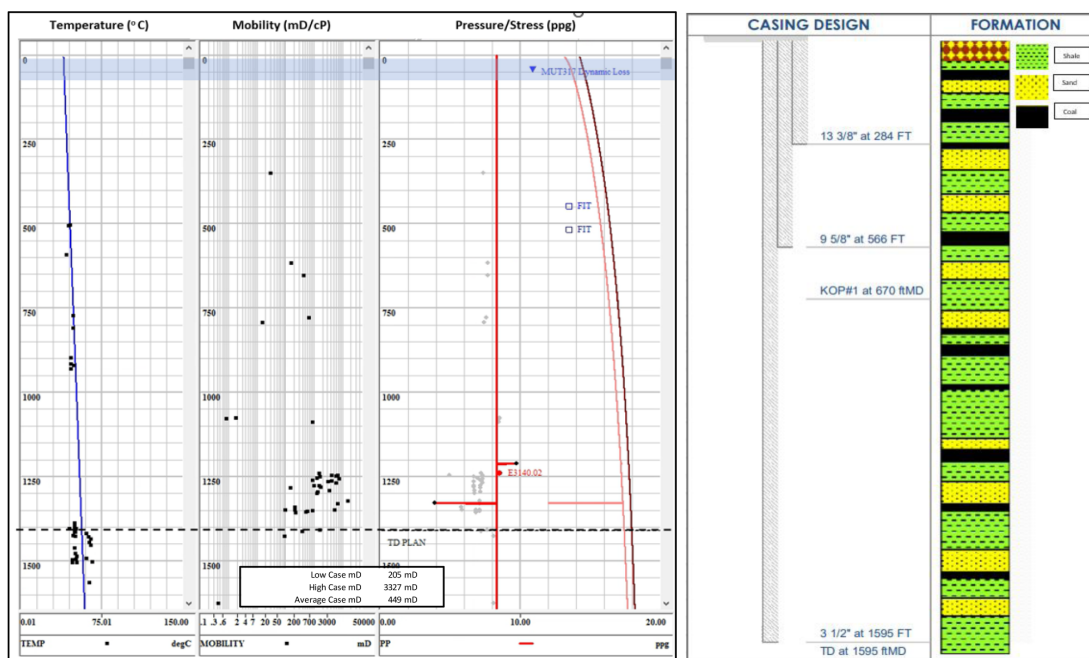


Figure 2—Pressure/stress profile and lithology of well "Px"

The designated well type is a "J" type, consisting of two sections with a maximum inclination of 15.65°. The overburden section was drilled using a 12¼" PDC bit, while an 8½" PDC bit was employed for drilling the reservoir section. A PDC bit generates large, firm, and intact cuttings by shearing the formation, in contrast to the gouging and crushing mechanism characteristic of a tricone bit. However, without effective shale inhibition, achieving such cuttings may not be possible, as reactive shales tend to disintegrate when exposed to inadequately inhibited water-based drilling fluids. The well schematic is provided in the figure below.

BIT & HOLE	WELL SCHEMATIC	CASING
Vertical Hole 12-1/4" Hole Section Primary Bit 12-1/4" PDC 566 ft MD 566 ft TVD 400 ft TVD SS	282 ft TVD 282 ft MD Interval:	13-3/8in Augered to 284 ftTVD/MD Casing spec: 9-5/8", 47 ppf, L-80, BTC, R3
Directional Hole 8-1/2" Hole Section Primary Bit 8-1/2" PDC Back up Bit 1,595 ft MD 1,576 ft TVD 1,410 ft TVD SS	1,010 ft TVD 1,029 ft MD Interval:	Perform FIT 13.5 PPG EMW Dual tubing completion 3-1/2", 9.2 ppf, L-80, SLA, R2

Figure 3—Well Schematic

Fluid System Development

Designing HPWBM is different from SOBM. A SOBM relies on a robust formulation and good filter cake qualities to drill a variety of challenging formations whereas the HPWBM would be designed specifically to the challenges in each particular formation. HPWBMs must be specifically tailored to the unique challenges of each drilling environment. As a result, HPWBF formulations vary significantly from one area to another depending on drilling challenges. A primary consideration in HPWBF design is the presence and characteristics of shale within the formation. Since no two shale formations are identical, it is imperative to customize the fluid system for each specific formation to ensure optimal performance and wellbore stability⁴. The development of the fluid system should be designed on a specific formation. Swelling, accretion and dispersion/erosion are the characteristics of clay that need to be designed for and tested during the design phase².

The designing processes consist of 2 steps that are laboratory tests and hydraulics simulation. The laboratory test consists of the shale study, physical and chemical properties and return permeability test. The Hydraulics simulations are conducted to understand the hydraulics behavior of the fluid. This hydraulics of the system was conducted with the drilling fluid provider proprietary simulation program. The simulation was conducted to understand the hole cleaning capability of the system as well as to simulate the Equivalent Circulating Density (ECD) of the system to avoid fracture induced fractures.

A comprehensive shale study should be done to understand the behavior of the rock in the specific area, then use the information of the laboratory result to select the composition of the fluid and the best formulation that demonstrates the most appropriate system. The shale study consists of the tests that are necessary to design the systems⁵:

- X-Ray Diffraction (XRD)
XRD is a basic tool in mineralogical analysis of shales. It is an analytical technique primarily used to determine a crystalline structure of materials and is the tool to identify the clay types in the sample.
- Cation Exchange Capacity (CEC)
CEC is a test to determine the quantity of reactive solids from a sample, based on the cation exchange capacity of the solids. Methylene blue solution is used to titrate a sample of drilled solids that have been dried, ground into a powder, then treated with a strong dispersant and oxidizing agent. A measured quantity of methylene blue dye is added and agitated vigorously. The dye is adsorbed by the reactive particles in the sample. When the active clay particles are saturated with adsorbed dye, based on their exchange capacity, the un-adsorbed dye appears and denotes the end point.
- Capillary Suction Time (CST)
The CST is a specialized filtration technique used to evaluate the dispersion and/or flocculation tendencies of shale formations when exposed to various components in base fluids. It is used for assessing the ionic inhibition capacity of a fluid, specifically its ability to suppress or prevent shale dispersion and water retention. The CST test quantitatively measures the degree to which fluid inhibits shale dispersion, thereby serving as an indicator of its shale-stabilizing properties.
- Shale Erosion Test
A Shale Erosion Test determines whether a drilling fluid exhibits erosion inhibition towards a particular shale/cutting sample. The known weight of dry sized shale is hot rolled in the test fluid (mud or brine) at 150°F for 16 hours. The shale sample is then washed, dried and weighed. The processed shale is then reported as a percentage of the original weight. This recovered weight is expressed as a percentage of the original weight. The greater the inhibition qualities of the mud or brine, the higher the shale recovery weight will be.

- Accretion Test

Shale Accretion Test determines the accretion and anti-agglomeration tendencies of drilling fluid. It measures the amount of shale adhered to a Monel tube after-rolling in the test fluid for 30 minutes at 150°F.

- Linear Swell Meter (LSM)

LSM measures the swelling tendency of shale in different solutions. A pulverized sample of shale is reconstituted by means of a hydraulic compactor unit. The cores thus produced are laterally confined in porous sleeves to minimizing radial swelling. The cores are placed in cups which are filled with the appropriate test fluid and the resultant linear swelling is recorded by a transducer and displayed graphically.

Table 1—Base Exchange Capacities of Clay Minerals (Grim/Weaver and Pollard).³

Clay Type	CEC (mEq / 100g)
Montmorillonite (Smectite)	70-100
Illite	10-40
Kaolinite	3-15
Chlorite	10-40
Attapulgite/sepiolite	10-35

On the Physical and Chemical properties test, the fluid was formulated and tested to achieve the pre-determined fluid properties target that align with the well condition. The fluids were formulated to be able to provide superior inhibition while at the same time provide proper hydraulics to be able to effectively drill within the ECD margin. A secondary focus was on the filtration capabilities, specifically on the Permeability Plugging Test (PPT). PPT is a test that evaluates the capability of the fluid to form the filter cake on the ceramic disk that simulates downhole condition. The PPT will determine the effectiveness of the solid in the drilling fluid to form a non-permeable filter cake against the reservoir.

The return permeability test or formation response test would be conducted prior to introducing the system into the reservoir to see how the actual rock interacts with the fluid. The result of this test will indicate whether fluid is damaging to the reservoir or not.

The drilling fluid was designed to meet four factors of HPWBF that are Highly Inhibitive, Non-Dispersed, Low Clay Content and shear thinning. Highly inhibitive characteristics were focused on how the fluid can inhibit and stabilize the shale formation. All issues regarding clay inhibition and stabilization such as swelling, dispersion and pore pressure transmission were accommodated in the design of this system. The non-dispersed characteristic could be achieved by not using any dispersant in the system along with controlled the pH in the system to not promote the dispersion. Low clay content in drilling fluid is achieved by diminishing the use of commercial clay in the system to minimize the ECD during drilling. The shear thinning of the mud would be achieved by using polymers for viscosifier and filtration control.

To initiate the solution creation, the XRD test was conducted to determine the mineral composition of the formation. Approximately 100 grams of cuttings samples from offset wells were dispatched to a third-party laboratory for XRD analysis. The XRD result shows the formation comprises three types of clay, Illite, Kaolinite and Smectite. Illite and Kaolinite are characterized as non-swelling clays, whereas Smectite exhibits swelling behavior upon hydration. The absorption of water molecules into the interlayer spaces of Smectite leads to an increase in its basal spacing. The Illite contains potassium (K^+) in its interlayer, which holds the layer together, stabilizes the clay structure and prevents swelling⁶. Although Illite does not swell, it can still adversely affect reservoir permeability. Under low-salinity or high-pH conditions, Illite

particles may detach and migrate, potentially causing pore throat plugging and a reduction in permeability within the reservoir formation. These effects are critical when designing water-based drilling fluids for Illite-rich formations⁷. Kaolinite is chemically stable under normal conditions but can undergo destabilization in high pH environments or due to potassium-induced illitization. This process transforms Kaolinite into Illite, altering its structure and making it more dispersible, thereby exacerbating wellbore instability⁸. Both swelling and dispersive clays can reduce reservoir permeability, leading to decreased production and potential wellbore instability. Therefore, water-based fluid systems should be carefully designed to mitigate these effects.

From the shale inhibition standpoint, the development of the fluid system began by looking at the base fluid design. Based on the XRD and CST results the base fluid formulation was built and tested using a combination of KCl and NaCl brine as the base brine. The combination of KCl and NaCl increases the osmotic pressure from the salinity of the fluid due to the difference in solute concentration across a semi-permeable membrane^{9, 10}. Higher salinity in the drilling fluid generates greater osmotic pressure, which limits the movement of water from the wellbore into the formation, a process that could otherwise lead to shale hydration and destabilization. When the formation has lower salinity than the drilling fluid, water tends to migrate from the formation toward the wellbore. The movement of water from the shale into the wellbore compresses the diffused double layer between clay platelets, thereby reducing shale permeability and enhancing membrane efficiency, which collectively contributes to improved shale stability.

Several formulations were evaluated based on the shale studies, including the HPWBF used in the upper section. The results of the shale study indicate that the formulation applied in the upper section demonstrated the most effective clay inhibition performance. Further adjustments were made to optimize the properties of the fluid to effectively address the challenges encountered in the reservoir section. Removal of flocculants and encapsulators minimizes the potential for formation damage associated with encapsulating polymers, which can adhere to grain surfaces and reduce permeability. A bridging agent was subsequently incorporated into the formulation to achieve low fluid loss, particularly in the Permeability Plugging Test (PPT), in conjunction with a nanotechnology-based sealant. Comparative testing revealed no significant performance difference between the fluid containing encapsulators and flocculants and the NHPWBF with respect to shale inhibition demonstrating effective clay stabilization even in the absence of these additives.

In addition to stabilizing the shale, another key objective in designing this fluid was to address well-specific challenges, particularly by minimizing filtrate invasion. This included a focused effort to reduce both spurt loss and total filtrate volume during the Pressure Point Test (PPT). Assuming the average permeability of the reservoir is 400 mD, a 20 μm ceramic disc was employed to simulate the pore throat characteristics of the reservoir. Additionally, the temperature and pressure conditions of the PPT were designed to replicate those of the reservoir environment. Excessive filtrate invasion can adversely impact the various lithologies present in the well. A significant fluid intrusion into the shale matrix can increase near-wellbore pore pressure over time, thereby reducing effective stress. This reduction in effective stress may lead to shear failure and ultimately compromise wellbore stability². Filtrate reduction can be achieved by sealing the microfractures within the shale formation. However, conventional filtration control additives were insufficient in meeting the targeted zero spurt loss in the PPT. Consequently, a recently developed nanotechnology sealant was incorporated into the formulation. This product was also expected to contribute to the stabilization of the coal formation by sealing the coal cleat system and provide a low filtrate loss to minimize the pressurization near wellbore cleat system that led to increase stress and instability³. The development of Nano Sealant products to stabilize shale microfractures has raised significant interest in oil and gas community¹². However, the common Nano Particle product was focused on stabilizing the impermeable formation such as shale. The new development hybrid organic/inorganic nanoparticle (HNP) sealant was introduced into the fluid system candidates during lab formulation. Unlike the common nano particles' products, this HNP product differs from conventional nanoparticles commonly used in drilling fluids, which are primarily designed to reduce filtrate loss in impermeable formations. Instead, it is capable

of reducing filtrate loss in both impermeable as well as permeable formations. HNP works with solids in the fluid system including filtration control agents, bridging agents, weighting agents and other solids¹². These characteristics of HNP expand its applicability in minimizing fluid invasion into the reservoir and reducing the risk of formation damage. The bridging agent was carefully designed with Wellbore Strength Enhancing Technology software from the service provider to match the reservoir pore throat.

Final Fluid Selection.

Based on the laboratory evaluation, the NHPWBF was selected. This system included the advanced HNP sealant designed to form a thin, impermeable filter cake on the wellbore wall on various types of formation. The NHPWBF met key performance criteria:

- Tight spurt loss (<1 mL)
- Low total fluid loss (< 8 mL in 30 minutes)
- Strong shale inhibition
- Thermal stability up to 200°F
- Minimal impact on reservoir permeability

The candidate fluid for reservoir consists of the following products:

Table 2—Fluid formulation

Product	NHPWBF
Name	Concentration (ppb)
Base Fluid	279.9
Alkalinity control	1
Filtration Control agents	9
HNP	5-9
Viscosifier	1
Salts	80
Clay Inhibitor	7.05
Anti-Bacterial	0.3
Bridging Package	11.25
Weighting Material	42.61

In the final formulation stage, the fluid system was re-tested to verify its inhibition properties—specifically swelling, accretion, and erosion—as well as its sealing capability, before being introduced into the reservoir. The shale inhibition's performance met expectations as shown in [table 3](#) below.

Sealing capability was tested with PPT assembly targeting the spurt volume below 1 mL and the total filtrate volume below 8 mL. The laboratory test shows that the target could be achieved as per design.

Table 3—NHPWBF Shale Study

Description	Water	NHPWBF
	Offset well	Offset well
Erosion	99.40%	3.50%
Accretion	N/A	0.20%
Swelling	6.90%	1.45%

Table 4—NHPWBF PPT result

NHPWBF Permeability Plugging Test		
Disc Size	20μ	
Pressure	1000psi	
Temperature	150°F	
Spurt	1.0mL	
Total Filtrate	6.0mL	

The HNP was tested at various concentrations, revealing that a higher concentration is required when the fluid is clean, whereas a lower concentration is sufficient when the fluid is contaminated with solids. Further adjustments to the formulation would be made at the rig site once drilling operations commenced.

The final evaluation of the fluid system, prior to determining whether the reservoir in this well could be drilled using NHPWBF as a replacement for the existing SOBM system—or to potentially continue with from the upper section—was the return permeability test. The core plug used in this test was obtained from a nearby offset well. This test was comparing the SOBM, NHPWBF and HPWBF as per table below.

Table 5—Return Permeability test comparison

Core List		Fluid Type	MW, ppg	Return Perm Recovery, %	Return Recovery (Scrape mud cake), %
Core Sample #1		SOBM	10.5	53	84
Core Sample #2		NHPWBF	10.5	48	81
Core Sample #3		HPWBF	10.5	44	78

The results indicated that recovery following exposure to the NHPWBF was superior to that of the conventional HPWBF and lower than that achieved with SOBM. Additional testing on the scraped mud cake was conducted to identify the location and extent of formation damage.

The image of the core following the return permeability test illustrates that fluid invasion is most extensive in the conventional HPWBF, compared to NHPWBF and SOBM. The circled areas indicate regions where filtrate did not penetrate the core. As shown in the image, the core exposed to NHPWBF contains more zones with limited filtrate invasion compared to SOBM, while the core treated with conventional HPWBF exhibits near-complete filtrate penetration throughout. No filtrate loss observed during dynamic and static testing in both NHPWBF and HPWBF samples, whereas the SOBM sample exhibited a filtrate volume of 1 mL during both dynamic and static testing processes. The return permeability tests were performed at reservoir condition of 200° F with fluid overbalance at 250 psi for 4 hours dynamic condition and 150 psi for 16 hours static condition.

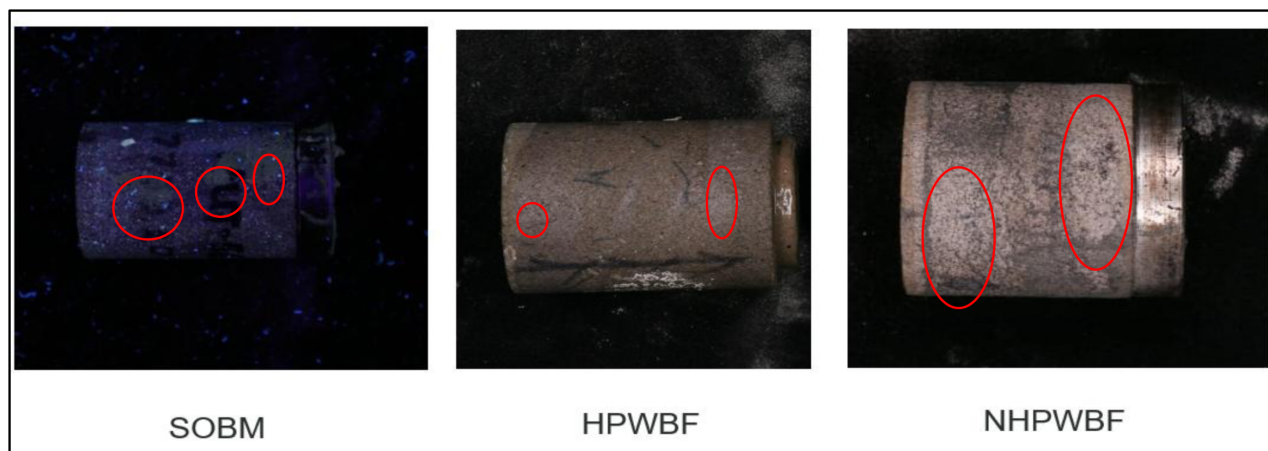


Figure 4—Core Appearance after Return Permeability Test

The operator expressed satisfaction with the recovery results observed during the return permeability test, as well as with the extent of fluid invasion into the core. A recovery exceeding 80 % on a tight, low-permeability core is considered a highly favorable outcome. Additionally, the test demonstrated a significant reduction in fluid invasion within the core.

Field Implementation

Drilling Strategy

The operator deployed the NHPWBF system in the reservoir section of the "P" well, while a conventional HPWBF was utilized in the overburden section. This strategy enabled cost optimization while enhancing formation protection in the pay zone. The HPWBF used in the overburden section is a standard fluid system commonly applied in this area. Drilling operations in the overburden section—whether using conventional techniques or the casing-while-drilling method—have generally proceeded smoothly, with no significant issues reported. The drilling fluid performance met expectations, with no issues related to shale inhibition. The cuttings condition during overburden drilling was firm with bit marks clearly observed.

The reservoir section was drilled using NHPWBF in accordance with the specified targets outlined below. The fluid design was carefully engineered to ensure effective hole cleaning, maintain borehole stability, and minimize formation damage. The hole cleaning strategy was developed using the fluid provider's proprietary hydraulics software to optimize rheology and drilling parameters, thereby maintaining the Equivalent Circulating Density (ECD) within the safe margin between pore pressure and fracture gradient.

Table 6—NHPWBF Target Properties

Properties	Unit	Target
Mud Weight	ppg	10.5 - 11
Rheology Temp.	°F	120
PV	cP	< 30
YP	lb/100ft ²	30 - 45
10 s Gel	lb/100ft ²	8 - 14
10 min Gel	lb/100ft ²	10 - 15
30 min Gel	lb/100ft ²	12 - 22
Tau0	lb/100ft ²	> 5
API filtrate	mL/30 min	< 4
HTHP Filtrate, 500 psi, 150°F	mL/30 min	< 6
pH		9.5 - 10.5
Methylene Blue Test	ppb	<7.5
Cl-	mg/L	120000 - 140000
Polyamine	wt%	2 - 3%
PPA, 20μ, 1000 psi, 150°F		
Spurt	mL	< 1
PPA Filtrate	mL	< 8

Drilling Performance

Drilling operations proceeded smoothly. No shale instability or clay swelling was observed, and firm cuttings throughout the reservoir section indicated effective shale inhibition. Coal layers were drilled without any sign of collapse or fluid loss events. No abnormal torque or drag was recorded, indicating good hole cleaning and lubricity.

Throughout the drilling operation, the rate of penetration (ROP) consistently remained high, with a peak ROP of 223 feet per hour (fph) on well "Px" and 307 fph on well "Py". The maximum Equivalent Circulating Density (ECD) recorded was 11.29 ppg, well below the fracture gradient of 17.51 ppg. No abnormal torque or drag was observed during the drilling process indicating that there was no cuttings accumulation in the wellbore and that the formation was effectively drilled and the wellbore cleaned.

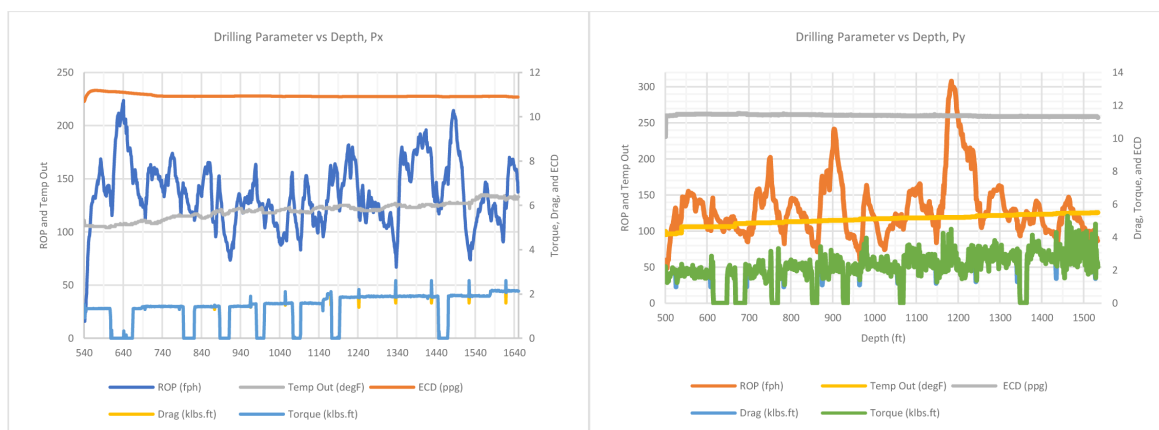


Figure 5—Drilling Parameter vs Depth

Drilling fluid properties were maintained to be within designed program throughout the drilling activity.

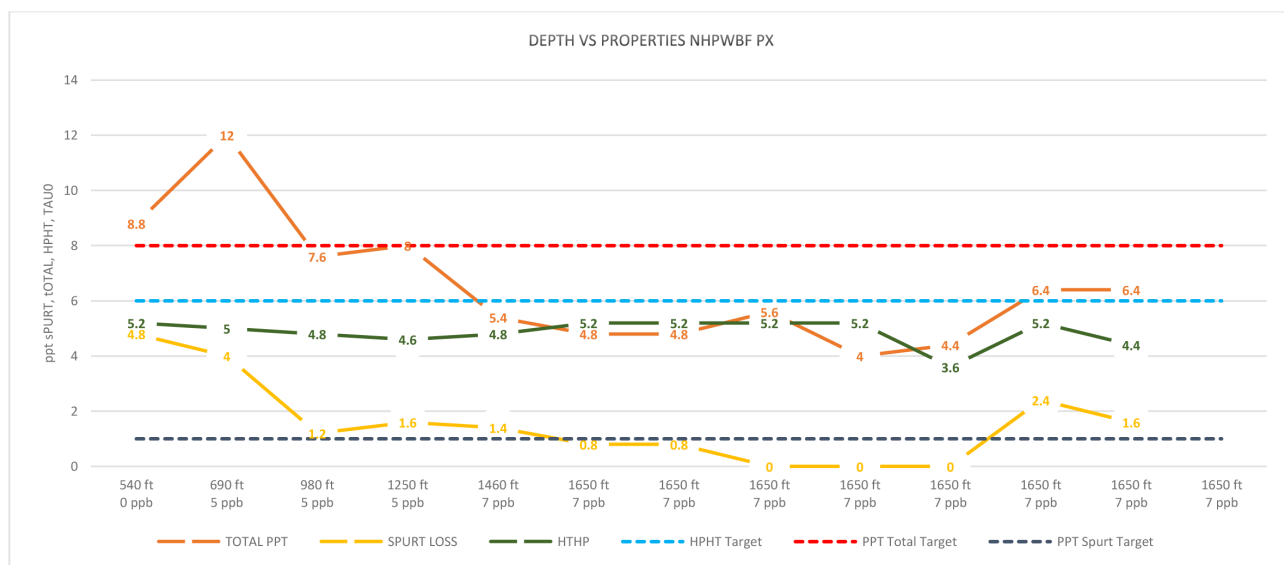


Figure 6—Depth vs Filtrate Properties

The initial plan was to utilize the HNP at its maximum concentration as determined through laboratory testing. However, the fluid was initially treated with a lower concentration and gradually increased until the target properties was reached. As a result, both spurt loss and total PPT filtrate were reduced to levels below the specified targets, with satisfactory performance achieved at a concentration at only 7 ppb.



Figure 7—Drilled Shale Cutting from 8 1/2" Reservoir Section

NHPWBF offers enhanced shale inhibition for the formation in this region. The image above illustrates effective inhibition, maintaining well-defined cuttings with clear bit marks. The cuttings remain well-preserved during transport from the bottom hole to the surface, experiencing minimal to no erosion, evidenced by sharp edges on both the cuttings and bit marks.

Wireline logging was conducted efficiently, with sampling times significantly lower than historical averages. All nine planned pressure tests and fluid analysis points were successfully executed. Each pressure point was acquired in under 20 minutes, with no indications of tool sticking or excessive fluid presence during the logging process. The pressure tests and fluid samples confirmed the integrity of the reservoir. Furthermore, the absence of drilling fluid filtrate in the downhole samples validated the sealing effectiveness of the nano sealant. In addition, the caliper sweep indicates an average borehole diameter of 8.75 inches, whereas the caliper log consistently records values between 8.5 and 9 inches, with ten data points reaching up to 10 inches. Three of these ten points are located within the coal layer, while the remaining points are

distributed across the sandstone and shale formations. This result was better to that of Offset P Well #2, which was drilled using SOBM and achieved an average hole size of 9.01 inches.

Economic Impact

The switch to NHPWBF yielded notable economic benefits. Drilling fluid costs were optimized, contributing to an overall expenditure reduction of 10.6% compared to wells drilled with SOBM. Furthermore, the minimized formation damage led to improved productivity. Two wells drilled with NHPWBF flowed naturally upon completion, a rare outcome in this mature field.

These outcomes demonstrate the economic and operational advantages of the NHPWBF system:

- Lower fluid costs

A comparison of the cost per barrel between two wells drilled with SOBM in the reservoir section (Offset #1 and Offset #2) and two wells drilled with NHPWBF in the reservoir section (Px and Py) indicates that the fluid cost was reduced by more than 60%.

- Reduced non-productive time

The operation was drilled faster and earlier than the planned drilling days, achieving 62% faster than the planned program

- Enhanced reservoir productivity

All wells drilled in this area previously required artificial lift to achieve the target production, even when SOBM was employed. However, the wells drilled using NHPWBF were able to produce through natural flow, eliminating the need for artificial lift. The production rates exceeded 500 barrels of oil per day (BOPD) with natural flow, surpassing initial expectations. This result demonstrates a significantly enhanced production performance and indicates the potential for additional cost savings.

- Compliance with environmental standards

The use of NHPWBF eliminates the need for cuttings handling associated with Oil on Cutting (OOC) treatment and the waste management requirements of SOBM, thereby reducing the risk of oil spillage during rig operations and the transportation of cuttings and sludge to final disposal sites. These waste management processes typically contribute to increased overall well costs, as this reduces fluid costs further by up to 25.28 % on post-drilling cutting treatment.

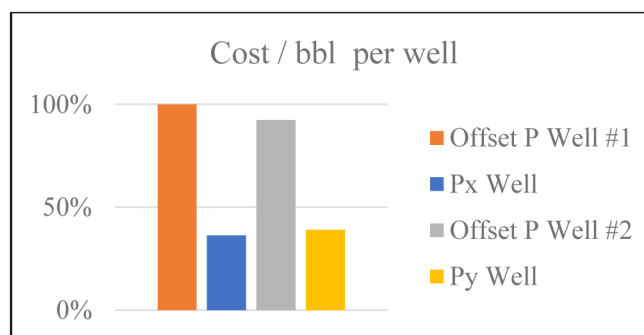


Figure 8—Cost/bbl comparison

Table 7—Planned drilling days vs execution days

Planned drilling days			Executed drilling days		% efficiency
12.25 section	0.8	days	0.2	days	25%
8.5 section	1.3	days	0.8	days	62%

Conclusion

Role of Nano Sealant

The key differentiator of the NHPWBF system was the incorporation of a newly developed HNP Sealant. This additive significantly improved filter cake quality, effectively reducing both spurt loss and total fluid loss, thereby minimizing fluid invasion into permeable formations.

The HNP sealant formed a non-permeable layer / mud cake at the wellbore-reservoir interface, preventing pore pressure transmission and stabilizing the shale formation.

Additionally, the sealant played a crucial role in stabilizing fragile coal cleat systems by sealing the coal cleat system and limiting fluid invasion, thus preventing sloughing, formation plugging, and potential wellbore collapse.

The Hybrid Nano-scale Sealant demonstrated effectiveness across various lithologies, including permeable formations such as sandstone and low-permeability formations such as shale, claystone, and coal layers.

Environmental Considerations

Unlike synthetic oil-based systems, the NHPWBF system is more environmentally benign. It poses lower disposal costs and reduces risk of contamination. Its formulation complies with environmental standards, supporting the operator's sustainability goals.

Reproducibility and Scalability

The success of the initial two wells demonstrated the reproducibility of the NHPWBF performance. The consistent results across different reservoir sections confirmed the robustness of the system. Plans are underway to extend its application field-wide, with necessary customizations based on reservoir characteristics.

The application of NHPWBF in East Kalimantan has proven to be a game-changing advancement in mature field drilling. The system successfully balanced wellbore stability, environmental compliance, and economic efficiency. Key conclusions include:

- The successful design and application of this system enabled natural flow in wells that typically require an artificial lift to achieve similar production.
- Drilling fluids costs were substantially reduced, thereby significantly lowering the overall cost of the well along with additional savings resulting from the elimination of artificial lift requirements. The total cost was significantly lower than the AFE.
- NHPWBF minimized formation damage through effective sealing and low filtrate invasion.
- The system provided excellent shale and coal stability.
- Operational efficiency improved, as evidenced by faster wireline logging and successful pressure testing.

Acknowledgments

The authors express gratitude to Pertamina Hulu Sanga Sanga, Indonesia and Halliburton Baroid for granting permission to publish this paper. The author acknowledges the Mud Engineers aboard the rig for their flawless execution of the mud system and extends appreciation to the laboratory team for their diligent efforts in preparing this project.

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